Fall 2025
ME 486/5586
Applied Finite Element Analysis

Final Group Project

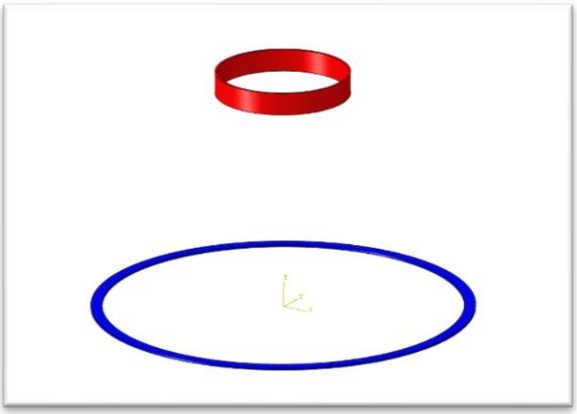
Your team has been formed to support new designs using Finite Element Method solutions. Your first application has arrived!

Your company produces Additive Manufacturing goods. You have been commissioned to create a new support stand for Tiny Houses. The support stands, to be made out of Aluminum, are to be constructed to withstand specific requirements. The mounting ring and ground base pad are given and should not be changed or altered. (The ground base pad should be the only part of the structure touching the ground). Simulations need to be performed to validate functional requirements.

- Axial Dead Load – 2000 N
- Lateral Load – 500 N
- Twisting Load – 100 N-mm
- Lowest Resonance – above 200 Hz
- Maximum Weight – below 120 N
- Mounting Schema – See below!

Your team will create 5 conceptual designs (in CAD) that minimize the weight concept, and then select just one for simulations and evaluations to see how it meets the current requirements.

The project requires (at a maximum) a 10-page summary as a report to your customer. The summary should be done in MS/Word format. In conjunction, a 5-10 minute presentation of the AM solutions to the Executives (instructor/students in class) is also required, with all team members presenting. The presentation should be in MS/PowerPoint format.



Detail mounting locations
Red – Mounting Ring
Blue – Ground Base Pad
(These are fixed and must be incorporated into your design).

Mounting designs are available in STEP and ACIS formats..